

## Calcic Xerosols

### Description:

Calcic Xerosols are mineral soils characteristic of arid and semi-arid climates, where evaporation exceeds precipitation.

They are distinguished by the presence of a **calcic horizon (Bk horizon)** within 50–100 cm of the surface, enriched with **secondary calcium carbonate ( $\text{CaCO}_3$ )**.

This horizon may appear as soft powdery lime, nodules, or hard calcrete layers (petrocalcic), depending on the degree of carbonate accumulation. These soils develop mainly on calcareous parent materials such as limestone, marl, or old alluvial deposits.

They are typically light-colored (pale brown to whitish), coarse to medium-textured, and low in organic matter. Their formation is linked to limited leaching, high evaporation, and the upward movement of soluble salts, which leads to the concentration of carbonates in the subsoil.

### Drainage:

- **Wet season:** Generally well-drained due to coarse textures and low clay content. However, infiltration may be restricted where a **hard calcrete layer** is close to the surface.
- **Dry season:** The soils become **very dry and hard**, often forming crusts at the surface that can inhibit seedling emergence. Moisture retention is **low** due to poor structure and limited organic matter.

### Landscape:

- Found on gently undulating plains, old alluvial fans, and limestone plateaus in semi-arid and arid regions. These landscapes typically receive 250–600 mm of rainfall annually.
- The soils occur extensively in parts of Eastern Kenya (Kitui, Makueni, Mwingi), Northern Kenya (Isiolo, Marsabit, Garissa), and Rift Valley lowlands (Baringo, Kajiado).

**Depth:**

Usually **moderately deep to deep (50–150 cm)**, but the effective rooting depth may be **restricted** by the presence of a **calcrete or stony layer**. In flatter areas, the soil may be deeper, while in eroded zones it can be shallow and gravelly.

**Workability:**

- **When wet:** Loose and friable; easy to cultivate; minimal stickiness.
- **When dry:** Becomes **hard, compact, and crusted**, making tillage difficult and reducing seed germination.
- **Overall:** Good workability after light rains or irrigation, but soil crusting and compaction can be problematic without organic amendments.

**Nutrient Richness & Cation Exchange Capacity (CEC):**

- **CEC:** Low to moderate (5–12 cmol(+)/kg), depending on clay and organic matter content.
- **Base saturation:** Very high (>80%) because of abundant calcium carbonate and other basic cations.
- **Nutrients:** Adequate calcium and magnesium, but **nitrogen, phosphorus, and zinc** are often deficient due to high pH and low organic matter.
- **Organic matter:** Low (<1.5%), leading to weak aggregation and limited biological activity.
- **Micronutrients:** Iron, manganese, and zinc deficiencies may occur under strong alkalinity.

**pH:**

Typically **alkaline (7.5–8.5)**. The high pH reduces the availability of phosphorus and micronutrients such as zinc and iron.

## Management:

- **Organic matter addition:** Incorporate farmyard manure, compost, or green manure to improve structure, water retention, and microbial activity.
- **Fertilization:** Apply nitrogen and phosphorus fertilizers; use acid-forming fertilizers such as **ammonium sulfate** to slightly reduce alkalinity.
- **Soil moisture management:** Adopt **water-harvesting techniques** (zai pits, contour bunds, terraces) to enhance moisture infiltration and storage.
- **Erosion control:** Maintain surface cover with mulch or vegetation to reduce wind and water erosion.
- **Irrigation:** Practice **controlled irrigation** to prevent salinization; ensure adequate drainage in irrigated systems.
- **Suitable crops:** Best suited for drought-tolerant species such as sorghum, millet, cowpeas, green grams, and pigeon peas. Under irrigation, crops like maize, tomatoes, onions, and watermelons can be grown successfully with proper fertility and salinity management.